ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

MID-SEM EXAMINATION — DECEMBER / JUNE 2024

B.TECH. CSE ENGINEERING — SEMESTER 4

Time Allowed: 1.5 Hours **Maximum Marks: 30**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- PART A ---

Q.1 Construct DFA accepting strings containing substring '010'. Convert NFA with epsilon to DFA. [15 Marks]

[OR]

Q.1 (Alt) State and prove Arden's Theorem. Find regular expression for given state diagram. [15 Marks]

--- PART B ---

Q.2 State and prove Pumping Lemma for Regular Languages. Prove L={0^n 1^n} is not regular. [15 Marks]

[OR]

Q.2 (Alt) Explain Chomsky Hierarchy. Construct Pushdown Automata for palindrome strings. [15 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****